

```
%-- 12/6/2024 12:22 AM --%
```

```
vortexShape
```

```
omega
```

```
vortexShape
```

```
omega
```

```
vortexShape
```

```
%-- 12/12/2024 9:39 AM --%
```

```
fftwork
```

```
load('30-300_2.8NF.mat')
```

```
fftwork
```

```
y = fft(velocity, N);
```

```
p2 = abs(y / N);
```

```
p1 = p2(1:floor(N / 2) + 1);
```

```
p1(2:end-1) = 2 * p1(2:end-1);
```

```
figure;
```

```
plot(f, p1.^2);
```

```
xlabel('Frequency (Hz)');
```

```
ylabel('Power');
```

```
title('Current Velocity Power Spectrum');
```

```
xlim([0 5000]);
```

```
spespectrogram(velocity, 256, 250, 256, Fs, 'yaxis');
```

```
spectrogram(velocity, 256, 250, 256, Fs, 'yaxis');
```

```
spectrogram(velocity, 512, 250, 256, Fs, 'yaxis');
```

```
figure;
```

```
spectrogram(velocity, 512, 250, 256, Fs, 'yaxis');
```

```
fc = 1000; % Cutoff frequency in Hz
```

```
lpFilt = designfilt('lowpassfir', 'PassbandFrequency', fc/(Fs/2), ...
```

```
'StopbandFrequency', (fc+100)/(Fs/2), 'PassbandRipple', 1, ...
```

```
'StopbandAttenuation', 60, 'DesignMethod', 'kaiserwin');
```

```
% Apply the low-pass filter to the velocity signal
```

```
filtered_velocity = filter(lpFilt, velocity);
```

```
figure;
```

```
spectrogram(filtered_velocity, 256, 250, 256, Fs, 'yaxis');
```

```
title('Spectrogram of Filtered Velocity Signal');
```

```
colorbar;
```

```
xlabel('Time (s)');
```

```
ylabel('Frequency (Hz)');
```

```
spectrogram(velocity, 512, 250, 256, Fs, 'yaxis');
```

```
figure;
```

```
spectrogram(velocity, 512, 250, 256, Fs);
```

```
fftwork
```

```
figure;
```

```
spectrogram(velocity, 512, 250, 256, Fs);
```

```
vnew = lowpass(velocity, 1000, 5000)
```

```
spectrogram(vnew, 512, 250, 256, Fs);
```

```
%-- 12/15/2024 1:07 PM --%
```

```
load('C:\Users\anneg\Documents\TPadWork\Finger Stretch Characterization\LDV ✓  
Data\LongerNiceChirps\chrip30-340.mat')
```

```
fftwork
```

```
%-- 12/17/2024 9:14 AM --%
```

```
load('C:\Users\anneg\Documents\TPadWork\Finger Stretch Characterization\LDV ✓  
Data\LongerNiceChirps\chrip30-340_NF_1.mat')
```

```
plot(temp)
```

```
dataNF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_1.mat");
dataNF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_2.mat");
dataF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_F_1.mat");
dataF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_F_2.mat");
dataNF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_1.mat");
dataNF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_1.mat");
dataNF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_2.mat");
dataF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_F_1.mat");
dataF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_F_2.mat");
clear
dataNF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_1.mat");
dataNF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_2.mat");
dataF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_F_1.mat");
dataF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_F_2.mat");
dataNF_1
dataNF_1.temp
close all
clear
dataNF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_1.mat");
plot(dataNF_1.temp)
close all
dataNF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_NF_2.mat");
plot(dataNF_2.temp)
dataF_1 = load("LDV Data\LongerNiceChirps\chirp30-340_F_1.mat");
plot(dataF_1.temp)
dataF_2 = load("LDV Data\LongerNiceChirps\chirp30-340_F_2.mat");
plot(dataF_2.temp)
fftwork
spectrogram(dataNF_1.temp, 512, 250, 256, Fs);
close all
spectrogram(dataNF_1.temp, 512, 250, 256, Fs);
fftwork
figure
spectrogram(velocityhp, 512, 250, 256, Fs);
spectrogram(velocityhp, 512, 250, 512, Fs);
figure
spectrogram(velocityhp, 512, 250, 256, Fs);
spectrogram(velocityhp, 512, 250, 128, Fs);
spectrogram(velocityhp, 512, 250, 512, Fs);
spectrogram(velocityhp, 512, 256, 512, Fs);
spectrogram(velocityhp, 512, 100, 512, Fs);
spectrogram(velocityhp, 512, 512, 512, Fs);
spectrogram(velocityhp, 512, 500, 512, Fs);
spectrogram(velocityhp, 256, 192, 512, 10000);
spectrogram(velocityhp, 1024, 768, 2048, 10000);
colormap jet
colormap parula
colormap hot
colormap cold
colormap grey
colormap gray
colormap turbo
colormap cool
```

```
colormap hsv
colormap bone
colormap copper
colormap sprign
colormap spring
colormap summer
colormap winter
colormap pink
colormap parula
colormap jet
colormap red
colormap pink
colormap jet
plot(temp)
plot(dataNF_1.temp)
fftwork
spectrogram(velocityhp, 1024, 768, 2048, 10000);
figure
spectrogram(velocityhp, 1024, 768, 2048, 10000);
spectrogram(temp, 1024, 768, 2048, 10000);
spectrogram(dataNF_1.temp, 1024, 768, 2048, 10000);
spectrogram(tempNF_1, 1024, 768, 2048, 10000);
signal = velocityhp
% Parameters
Fs = 10000; % Sampling frequency
Fc = 1000; % Lowpass cutoff frequency
windowLength = 1024; % Spectrogram window length
overlap = 768; % 75% overlap
fftPoints = 2048; % Number of FFT points
% Generate Spectrogram
[s, f, t] = spectrogram(signal, windowLength, overlap, fftPoints, Fs);
% Frequency Range for THD Analysis
freqCenters = 10:20:350; % Frequency centers (10, 30, 50, ..., 350 Hz)
bandWidth = 10; % Half-width of each band (±10 Hz around center)
% Initialize THD Results
thdValues = [];
% Compute THD for Each Frequency Band
for fc = freqCenters
    % Define frequency range for the current band
    bandIdx = (f >= (fc - bandWidth)) & (f <= (fc + bandWidth));
    % Extract power within the frequency band
    bandPower = abs(s(bandIdx, :)).^2;
    totalPower = sum(bandPower, 1);
    % Compute THD: Ratio of harmonics power to total power
    thd = 100 * (1 - (max(bandPower, [], 1) ./ totalPower)); % Example THD formula
    thdValues = [thdValues; thd];
end
% Box Plot of THD Across Frequency Bands
figure;
boxplot(thdValues', freqCenters);
xlabel('Frequency (Hz)');
ylabel('THD (%)');
title('THD Across Frequency Bands');
```

```
grid on;
clear
fftwork
signal = velocityhp
% Parameters
Fs = 10000; % Sampling frequency
Fc = 1000; % Lowpass cutoff frequency
windowLength = 1024; % Spectrogram window length
overlap = 768; % 75% overlap
fftPoints = 2048; % Number of FFT points
% Generate Spectrogram
[s, f, t] = spectrogram(signal, windowLength, overlap, fftPoints, Fs);
% Frequency Range for THD Analysis
freqCenters = 10:20:350; % Frequency centers (10, 30, 50, ..., 350 Hz)
bandWidth = 10; % Half-width of each band (±10 Hz around center)
% Initialize THD Results
thdValues = [];
% Compute THD for Each Frequency Band
for fc = freqCenters
    % Define frequency range for the current band
    bandIdx = (f >= (fc - bandWidth)) & (f <= (fc + bandWidth));
    % Extract power within the frequency band
    bandPower = abs(s(bandIdx, :)).^2;
    totalPower = sum(bandPower, 1);
    % Compute THD: Ratio of harmonics power to total power
    thd = 100 * (1 - (max(bandPower, [], 1) ./ totalPower)); % Example THD formula
    thdValues = [thdValues; thd];
end
% Box Plot of THD Across Frequency Bands
figure;
boxplot(thdValues', freqCenters);
xlabel('Frequency (Hz)');
ylabel('THD (%)');
title('THD Across Frequency Bands');
grid on;
THDBox
spectrogram(velocityhp, 1024, 768, 2048, 10000);
THDBox
close all
fftwork
plot(temp)
plot(tempNF_1)
figure
plot(tempNF_1)
fftwork
%-- 1/8/2025 8:21 PM --%
load('C:\Users\anneg\Documents\TPadWork\Finger Stretch Characterization\LDV✓
Data\LongerNiceChirps\chirp30-340_F_2.mat')
fftwork
spectrogram(velocityhp, 1024, 768, 2048, 10000);
THDBox
signal = velocityhp
THDBox
```

```
fftwork
spectrogram(velocityhp, 1024, 768, 2048, 10000);
clear
spectrogram(velocityhp, 1024, 768, 2048, 10000);
fftwork
figure
spectrogram(velocityhp, 1024, 768, 2048, 10000);
fftwork
plot(velocity)
figure
plot(velocity)
fftwork
fftwork
clear
fftwork
plot(useTemp)
figuer
figure
plot(useTemp)
fftwork
figure
plot(useTemp)
%-- 1/9/2025 9:28 AM --%
dataNF_1 = load("LDV Data\chirp30-340_NF_1.mat");
dataNF_1 = load("LDV Data\chirp30-340_NF1.mat");
dataNF_2 = load("LDV Data\chirp30-340_NF2.mat");
dataNF1 = load("LDV Data\chirp30-340_NF1.mat");
dataNF2 = load("LDV Data\chirp30-340_NF2.mat");
dataNF3 = load("LDV Data\chirp30-340_NF3.mat");
dataF1 = load("LDV Data\chirp30-340_F1.mat");
dataF2 = load("LDV Data\chirp30-340_F2.mat");
dataF3 = load("LDV Data\chirp30-340_F3.mat");
tempNF1 = dataNF1.temp;
tempNF2 = dataNF2.temp;
tempNF3 = dataNF3.temp;
tempF1 = dataF1.temp;
tempF2 = dataF2.temp;
tempF3 = dataF3.temp;
plot(tempNF1)
close all
plot(tempNF2)
close all
plot(tempNF3)
close all
plot(tempNF1)
plot(tempNF2)
close all
plot(tempNF3)
plot(tempF1)
plot(tempF2)
plot(tempF3)
plot(tempF1)
fftwork
```

```
figure
useRanges(1)
clear
fftwork
figure
plot(useTemp(useRanges
plot(useTemp(useRanges))
plot(useTemp(useRanges(1)))
useRanges(1)
plot(useTemp(useRanges{1}))
plot(tempNF1)
fftwork
plot(tempNF2)
figure
plot(tempNF2)
fftwork
plot(useTemp(useRanges{1}))
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
fftwork
figure
plot(useTemp(useRanges{1}))
plot(useTemp(useRanges{2}))
plot(useTemp(useRanges{1}))
plot(useTemp(useRanges{2}))
plot(tempNF2)
fftwork
figure
plot(useTemp(useRanges{1}))
plot(useTemp(useRanges{2}))
plot(useTemp(useRanges{3}))
plot(useTemp(useRanges{4}))
plot(useTemp(useRanges{5}))
plot(useTemp(useRanges{6}))
plot(useTemp(useRanges{7}))
plot(useTemp(useRanges{8}))
plot(useTemp(useRanges{9}))
plot(useTemp(useRanges{10}))
plot(useTemp(useRanges{11}))
fftwork
spectrogram(useTemp(useRanges{1}))
```

```
spectrogram(useTemp(useRanges{1}), 1024, 768, 2048, 10000);
spectrogram(useTemp(useRanges{1}), 1024, 768, 2048, 5000);
spectrogram(useTemp
spectrogram(useTemp, 1024, 768, 2048, 5000);
useTemp = lowpass(useTemp,1000, Fs);
spectrogram(useTemp, 1024, 768, 2048, 5000);
fftwor
spectrogram(useTemp, 1024, 768, 2048, 5000);
figure
plot(useTemp(useRanges{1}))
288-150
fftwor
plot(useTemp(useRanges{1}))
figure
plot(useTemp(useRanges{1}))
78065-200
103167-200
119047-100
190084-200
fftwor
spectrogram(useTemp, 1024, 768, 2048, 5000);
spectrogram(useTemp(useRanges{1}), 1024, 768, 2048, 5000);
fftwor
figure
plot(useTemp(useRanges{1}))
spectrogram(useTemp(useRanges{1}), 1024, 768, 2048, 5000);
spectrogram(useTemp(useRanges{2}), 1024, 768, 2048, 5000);
spectrogram(useTemp(useRanges{3}), 1024, 768, 2048, 5000);
fftwor
figure
plot(useTemp(useRanges{1}))
plot(useTemp(useRanges{2}))
figure
plot(useTemp(useRanges{2}))
fftwor
title("aas")
title("No Finger Spectrogram")
colormap(jet)
colormap(default)
parula
colormap(parula)
fftwor
xticks(0:0.25:max(f/1000));
xlim(
xlim
help xlim
xlim(0,1500)
xlim(0:1.25)
xlim(0:1.5)
title("No Finger Spectrogram")
yticks(0:0.25:3)
yticks(0:0.25:3.25)
fftwor
```

```
title("Spectrogram of Average Velocity Signal (IFFT)")
yticks(0:0.25:3.25)
xlim(0:1.5)
xticks(0:0.25:max(f/1000));
THDTutorial
figure;
fftworke
close all
window = hamming(256)
clc
window = hamming(256);
fftworke
figure
% Spectrogram
noverlap = 128;
nfft = 1024;
spectrogram(useTemp(useRanges{curr}), window, noverlap, nfft, Fs);
fftworke
[s, f, t, p] = spectrogram(useTemp(useRanges{curr}), window, noverlap, nfft, Fs);
avg_power = mean(p, 2);
magnitude = sqrt(avg_power);
frequencies = f;
fftworke
THDBox
%-- 1/10/2025 9:03 AM --%
fftworke
close all
%-- 1/14/2025 7:27 PM --%
dataBox = load("LDV Data\chirp30-340_F3.mat");
plot(dataBox.temp)
clear
dataBox = load("LDV Data\chirp30-340_Box.mat");
plot(dataBox.temp)
fftworke
close all
figure
fftworke
close all
clear
fftworke
close all
spectrogram(useTemp, window, noverlap, nfft, Fs);
spectrogram(useTemp, window, 768, nfft, Fs);
spectrogram(useTemp, window, 500, nfft, Fs);
spectrogram(useTemp, window, 256, nfft, Fs);
spectrogram(useTemp, window, 25, nfft, Fs);
close all
figure
spectrogram(useTemp, window, 255, nfft, Fs);
spectrogram(useTemp, window, 128, 2048, Fs);
%-- 1/15/2025 10:04 AM --%
G1 = zpk([-3],[-1 -2], 3)
G2 = tf([3 9],[1 3 2])
```



```
G1==G2
clc
G1
G2
G1*G2
G1
G2
clc
G1-G2<<
G1-G2<<2
clc
help zpk
help tf
H1 = feedback(G1,G2)
H1 = feedback(G1,G2, -1)
simplify
simp
sim
%-- 1/15/2025 10:22 AM --%
G1 = zpk([-3],[-1 -2], 3)
G2 = tf([3 9],[1 3 2])
feedback(G1,G2)
H1 = feedback(G1,G2)
step(H1)
step(G3/(s+1)^3)
step(G2/(s+1)^3)
s = tf([1][])
s = tf([1 0][1])
s = tf([1 0],[1])
step(H1/(s+1)^3)
clc
%-- 1/16/2025 9:13 AM --%
fftwork
close all
fftwork
close all
clear
teensySinWaves
fftwork
fdesign.abstracttypewspecs.syncspecs
fftwork
close all
fftwork
%-- 1/16/2025 1:23 PM --%
fftwork
xcorr(useTemp)
clear
fftwork
figure
xcorr(useTemp);
[c,lags] = xcorr(useTemp);
stem(lags,c)
fftwork
```

```
[c,lags] = xcorr(useTemp,'normalized');
[c,lags] = xcorr(useTemp,'normalized');
[c,lags] = xcorr(useTemp,'normalized');
stem(lags,c)
[c,lags] = xcorr(useTemp);
stem(lags,c)
plot(useTemp)
figure
stem(lags,c)
correlation
length(signal)
correlation
length(signal)
correlation
correlation
correlation
correlation
correlation
correlation
%-- 1/17/2025 11:05 AM --%
correlation
%-- 1/24/2025 3:53 AM --%
fftwork
length(useTemp(useRanges{1}))
length(useTemp(useRanges{1}))
a= useTemp(useRanges{1});
plot(a)
length(a)
L = ans
t5 = tukeywin(L,0.5)
plot(t5)
%-- 1/24/2025 12:19 PM --%
fftwork
clear
correlation
fftwork
figure(9)
colormap(jet)
colormap(blue)
colormap(ble)
help colormap
help graph3d
fftwork
clear
close all
%-- 1/30/2025 11:32 AM --%
signal_generator
sendSineWithEnvelope([], 100, 0.8, 3, 10000, 0.5, 0); % Plot
sendChirpWithEnvelope([], 30, 350, 0.8, 2, 10000, 0.3, 0); % Plot
sendSineWithEnvelope([], 100, 0.8, 3, 10000, 0.5, 0); % Plot
sendChirpWithEnvelope([], 30, 350, 0.8, 2, 10000, 0.3, 0); % Plot
signal_generator
sendChirpWithEnvelope
signal_generator
clear
```

```
signalGen
signals = signalGen();
signals.sinWithEnvelope([], 100, 0.8, 3, 10000, 0.5, 0);
signals.sineWithEnvelope([], 100, 0.8, 3, 10000, 0.5, 0);
signals.sineWithEnvelope([], 100, 0.8, 5, 10000, 0.5, 0);
signalGen
signals.sineWithEnvelope([], 100, 0.8, 5, 10000, 0.5, 0);
signalGen
clear
signals = signalGen();
signals.sineWithEnvelope([], 1, 0.8, 3, 10000, 0);
signals.sineWithEnvelope([], 100, 0.8, 3, 10000, 0);
signals.sineWithEnvelope([], 1, 1, 5, 10000, 0);
signals.sineWithEnvelope([], 2, 1, 5, 10000, 0);
signals.sineWithEnvelope([], 30, 1, 5, 10000, 0);
signals.sineWithEnvelope([], 100, 1, 5, 10000, 0);
signals.sineWithEnvelope([], 50, 1, 2, 10000, 0);
signals.sineWithEnvelope([], 50, 1, 0.5, 10000, 0);
tukeyAlpha
0.5*0.25/0.5
tukeywin(length(100),0.25)
plot(tukeywin(length(100),0.25)
plot(tukeywin(length(100),0.25))
fs = 1000; % Sampling frequency (Hz)
duration = 2; % Signal duration in seconds
N = fs * duration; % Number of points
tukeyAlpha = 0.5; % Tukey window shape parameter (0 = rectangular, 1 = Hann)
window = tukeywin(N, tukeyAlpha); % Generate Tukey window
t = linspace(0, duration, N); % Time vector
figure;
plot(t, window, 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Amplitude');
title(['Tukey Window (\alpha = ', num2str(tukeyAlpha), ')']);
grid on;
fs = 1000; % Sampling frequency (Hz)
duration = 2; % Signal duration in seconds
N = fs * duration; % Number of points
tukeyAlpha = 0.25; % Tukey window shape parameter (0 = rectangular, 1 = Hann)
window = tukeywin(N, tukeyAlpha); % Generate Tukey window
t = linspace(0, duration, N); % Time vector
figure;
plot(t, window, 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Amplitude');
title(['Tukey Window (\alpha = ', num2str(tukeyAlpha), ')']);
grid on;
fs = 1000; % Sampling frequency (Hz)
duration = 1; % Signal duration in seconds
N = fs * duration; % Number of points
tukeyAlpha = 0.25; % Tukey window shape parameter (0 = rectangular, 1 = Hann)
window = tukeywin(N, tukeyAlpha); % Generate Tukey window
t = linspace(0, duration, N); % Time vector
```

```
figure;
plot(t, window, 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Amplitude');
title(['Tukey Window (\alpha = ', num2str(tukeyAlpha), ')']);
grid on;
signals.sineWithEnvelope([], 50, 1, 0.5, 10000, 0);
fs = 1000; % Sampling frequency (Hz)
duration = 0.5; % Signal duration in seconds
N = fs * duration; % Number of points
tukeyAlpha = 0.25; % Tukey window shape parameter (0 = rectangular, 1 = Hann)
window = tukeywin(N, tukeyAlpha); % Generate Tukey window
t = linspace(0, duration, N); % Time vector
figure;
plot(t, window, 'LineWidth', 1.5);
xlabel('Time (s)');
ylabel('Amplitude');
title(['Tukey Window (\alpha = ', num2str(tukeyAlpha), ')']);
grid on;
signals.sineWithEnvelope([], 50, 1, 0.5, 10000, 0);clear
clear
close all
clc
clear
signalGen
signals = signalGen();
clear
signals = signalGen();
signalGen
signals = signalGen();
signals.sineWithEnvelope([], 5, 1, 0.5, 10000, 0);
signals.sineWithEnvelope([], 30, 1, 0.5, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 0.5, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 0.5, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 2, 10000, 0);
signals.sineWithEnvelope([], 30, 1, 1, 10000, 0);
signalGen
clear
signals = signalGen();
signals.sineWithEnvelope([], 30, 1, 1, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 1, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 1, 10000, 0);
signalGen
signals.sineWithEnvelope([], 30, 1, 1, 10000, 0);
signals.chirpWithEnvelope([], 30, 350, 0.8, 2, 10000, 0);
signalGen
signals.chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
signalGen
```

```
signals.chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
signalGen
signals.chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
fftwork2
signalGen
clear
clc
a = signalGen();
a.chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
[b,p] = chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
[b,p] = a.chirpWithEnvelope([], 30, 350, 1, 1, 10000, 0);
% Generate a sine wave with envelope
[funcs] = signalGen(); % Get the signal generation functions
% Define parameters for the sine wave
freq = 50;           % Frequency in Hz
amplitude = 1;       % Amplitude in volts
duration = 1;        % Duration in seconds
fs = 10000;          % Sampling frequency in Hz
% Call the function to generate the sine wave with envelope
[sineSignal, t] = funcs.sineWithEnvelope(freq, amplitude, duration, fs);
% The signal is now stored in 'sineSignal' and time in 't'
% You can use it for further processing, plotting, or sending to hardware
% The plot will also show automatically due to the plotSignal call inside the function
% Generate a chirp signal with envelope
freqStart = 30;      % Starting frequency in Hz
freqEnd = 350;       % Ending frequency in Hz
amplitude = 1;       % Amplitude in volts
duration = 1;        % Duration in seconds
% Call the function to generate the chirp signal with envelope
[chirpSignal, t] = funcs.chirpWithEnvelope(freqStart, freqEnd, amplitude, duration, fs);
% The chirp signal is now stored in 'chirpSignal' and time in 't'
% It will also be plotted automatically
fftwork2
clc
fftwork2
figure 3
figure 3;
figure(3)
help xlim
xlim([0 300])
figure(4)
xlim([0 400])
xlim([0 800])
xlim([0 1000])
fftwork2
lear
clear
clc
fftwork2
fftwork3
interpolatedImpedance
%-- 1/30/2025 6:33 PM --%
fftwork2
```

```
EE360_HW1
clear
close all
EE360_HW1
clear
clc
interpolatedImpedance
fftworke2
fftworke
%-- 2/7/2025 5:12 AM --%
sineWaveDeconv
figure
x = chirp(t, f1, T, f2, 'linear')
plot(x)
sineWaveDeconv
close all
sineWaveDeconv
figure 3
figure(3)
sineWaveDeconv
%-- 2/14/2025 2:31 PM --%
TestLDV
license
%-- 2/14/2025 3:19 PM --%
TestLDV
daqvendorlist
TestLDV
daqvendorlist
TestLDV
daq.getVendors()
daqreset
daq.getVendors()
%-- 2/18/2025 2:07 PM --%
load('interpolatedImpedance_data.mat')
print("hi")
%-- 2/18/2025 2:37 PM --%
sineWaveDeconv
%-- 2/18/2025 3:32 PM --%
sineWaveDeconv
fftworke3
fftworke
%-- 2/20/2025 12:37 AM --%
tests
ear serialObj % Clears the object from memory
delete(serialportfind)
clear serialObj
delete(serialportfind)
tests
clear serialObj
delete(serialportfind)
tests
sendToTeensy
tests
```

```
clear serialObj % Clears the object from memory
delete(serialportfind) % Ensures no lingering serial connections
tests
signalGen
tests
clear serialObj % Clears the object from memory
delete(serialportfind) % Ensures no lingering serial connections
tests
sendToTeensy
tests
sendToTeensy
tests
sendToTeensy
tests
sendToTeensy
tests
clear serialObj % Clears the object from memory
delete(serialportfind) % Ensures no lingering serial connections
tests
sendToTeensy
tests
untitled3
t100khz
tests
signalGen
tests
THDTutorial
tests
clear
tests
close all
tests
sineWaveDeconv
%-- 2/20/2025 7:29 PM --%
correlation
tests
clear
close al
close all
tests
close all
tests
close all
tests
close all
%-- 2/22/2025 11:07 AM --%
tests
tests
tests
close all
tests
TestLDV
%-- 2/22/2025 11:26 AM --%
```

```
TestLDV
Data.dq.Dev1_ai11
Data.dq.Dev1
Data.dq
Data_dq
Data_dq.Time
Data__dq.Dev1_ai15
Data_dq.Dev1_ai15
clc
TestLDV
tests
>> tempVec(1)
tempVec
TestLDV
tests
tempVec
tests
disp(Data_dq.Properties.VariableNames)
tests
daq.getDevices
daqSession.AvailableChannels
tests
%-- 2/27/2025 1:17 PM --%
load('trials.mat')
plot(tempVec)
tempVec
tempVec(1)
plot(tempVec(1))
tempVec{1, 1}
plot(tempVec{1, 1})
plot(tempVec{1, 1})
temp = tempVec{1,1}
tempLP = lowpass(temp, 1000,5000)
plot(tempLP)
plot(temp)
close all
figure
plot(temp)
hold on
plot(tempLP)
window = hamming(256)
noverlap = 128
nfft = 1025
figure;
spectrogram(tempLP, window, noverlap, nfft, Fs);
Fs = 5000
spectrogram(tempLP, window, noverlap, nfft, Fs);
colormap(jet)
temp
tempLP = tempLP - mean(tempLP)
close all
figure;
plot(temp)
```



```
hold on
plot(tempLP)
b = fir1(700, 11/(Fs/2), 'high')
tempHPLP = filter(b, 1, tempLP);
plot(tempHPLP)
figure
plot(tempHPLP)
spectrogram(tempHPLP, window, noverlap, nfft, Fs);
N = length(tempHPLP)
y = fft(tempHPLP, N);
p2 = abs(y/N);
p1 = p2(1:floor(N/2)+1);
p1(2:end-1) = 2* p1(2:end-1);
figure;
f = Fs * (0:(floor(N/2)+1, 1);
f = Fs * (0:(floor(N/2)+1), 1);
f = Fs * (0:floor(N/2)+1, 1);
f = Fs * (0:(floor(N / 2))) / N;
close all
plot(f, p1.^2);
xlim([0 1000])
xlim([0 500])
close all
clc
figure
plot(temp)
plot(temp)
hold on
plot(tempLP)
plot(tempHPLP)
clear
clc
load('trials.mat')
temp = tempVec{1,1}
plot(temp)
close all
clc
plot(temp)
%-- 2/28/2025 4:52 PM --%
-0.0184(0.0068)
-0.0184*(0.0068)
/(0.26478*0.0063)
-0.0184*(0.0068)/(0.26478*0.0063)
sssss
%-- 2/28/2025 5:42 PM --%
sssss
close all
pb
rlocus(Gpsi)
pb
rlocus(Gpsi);
close all
rlocus(Gpsi);
```

```
Gpsi
gak = 1/Gpsi
s0=-0.13
evalfr(gak,s0)
rlocfind
[K, poles] = rlocfind(Gpsi);
K
[K, poles] = rlocfind(Gpsi);
K
[K, poles] = rlocfind(Gpsi);
K
[K, poles] = rlocfind(Gpsi);
K
K = 1.56
[K, poles] = rlocfind(Gpsi);
K
[K, poles] = rlocfind(Gpsi);
K
pb
sssss
pb
close all
rloc
rlocus(Gpsi)
rlocfind(Gpsi)
K
[K, poles] = rlocfind(Gpsi);
poles
K
[K, poles] = rlocfind(Gpsi);
K
poles
[K, poles] = rlocfind(Gpsi);
poles
[K, poles] = rlocfind(Gpsi);
poles
K
pb
close all
p2
close all
p2
close all
p2
G
D
p2
close all
p2
close all
% Define the plant transfer function G(s)
s = tf('s');
G = 10 / (s * (s + 1) * (s + 10)); % Given plant transfer function
```

```
% Define the lead compensator  $D_{\text{lead}}(s) = K * (s + z) / (s + p)$ 
K_lead = 114; % From previous lead compensator design
z_lead = 0.9;
p_lead = 22.5;
D_lead = K_lead * (s + z_lead) / (s + p_lead);
% Define the lag compensator  $D_{\text{lag}}(s) = K_{\text{lag}} * (s + z) / (s + p)$ 
K_lag = 4444.4; % From lag compensator design
z_lag = 0.1; % Chosen zero for the lag compensator
p_lag = 10; % Chosen pole for the lag compensator
D_lag = K_lag * (s + z_lag) / (s + p_lag);
% Combine the lead and lag compensators
D_total = D_lead * D_lag;
% Define the open-loop transfer function  $L(s) = D(s) * G(s)$ 
L = D_total * G;
% Calculate the closed-loop transfer function with unity feedback
T = feedback(L, 1);
% Plot the step response of the closed-loop system
figure;
step(T);
title('Step Response of the Closed-Loop System');
p2
Dlead
p2
close all
p2
49/4.56
p2
close all
p2
p4
E
clear K
p4
G
p4
P
p4
P
p4
theta
p4
close all
p4
[K, poles] = rlocfind(OL);
[K, poles] = rlocfind(theta);
p4
close all
p4
sgrid(zeta, []);
sgrid(s, []);
clear
p4
sgrid([]);
```

```
p4
close all
p4
close all
p4
close all
p4
close all
p4
clear
p4
clear
close all
p4
close all
p4
sgrid
p4
sssss
close all
p2
p4
%-- 3/6/2025 5:31 PM --%
tests
%-- 3/7/2025 4:02 AM --%
tests
plot(tvec{1},tempVec{1})
tests
load('30-350HzChirpEnv0.8.mat')
plot(tvec{1},tempVec{1})
close all
plot(tvec{1},tempVec{1})
tests
plot(tvec{1},tempVec{1})
tests
close all
plot(tvec{1},tempVec{1})
tests
plot(tvec{1},tempVec{1})
tests
t1 = tempVec{1}
load('30-350HzChirpEnv0.8.mat')
t1 = tempVec{1};
window = hamming(256);
noverlap = 128;
nfft = 1024;
Fs
Fs = 10000
close all
figure;
-0.0220
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-0.0180
```

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load('30-350HzChirpEnv0.8.mat')
t1 = tempVec{1};
window = hamming(256);
noverlap = 128;
nfft = 1024;
Fs
```

```
Unrecognized function or variable 'Fs'.
Fs = 10000
Fs =
10000
spectrogram(t1, window, noverlap, nfft, Fs);
figure;
t1
spectrogram(t1, window, noverlap, nfft, Fs);
t1 = lowpass(t1, 1000, Fs);
spectrogram(t1, window, noverlap, nfft, Fs);
t2 = lowpass(t1, 1000, Fs)
close all
spectrogram(t2, window, noverlap, nfft, Fs);
load('trials.mat')
plot(tVec{1}, tempVec{1})
plot(tvec{1}, tempVec{1})
load('10HzSin3.3.mat')
plot(tvec{1}, tempVec{1})
tests
plot(tvec{1}, tempVec{1})
fftwork3
fftwork2
fftwork3
postProcessFreqResponse
t1 = tempVec{1};
tempVec
load('30-350HzChirpEnv0.8.mat')
load('30-350HzChirpEnv0.7.mat')
t1 = tempVec{1};
plot(t1)
temp = t1;
postProcessFreqResponse
temp = t1(4512:30492);
postProcessFreqResponse
load('30-350HzChirpEnv0.7.mat')
impulseRecovery
close all
impulseRecovery
close all
plot(tempVec{1})
hold on
plot(tempVec{2})
plot(tempVec{3})
figure
plot(tempVec{3}(4500:30800))
30800-4500
plot(chirpTest);
plot(tempVec{1})
plot(chirpTest)
impulseRecovery
4500+2500
4500+25000
plot(tempVec{1}(4550:29550))
```

```
plot(tempVec{1}(4550:29550))
plot(tempVec{1}(5000:30000))
plot(tempVec{1}(5050:30050))
impulseRecovery
close all
impulseRecovery
close all
impulseRecovery
load('30-350HzChirpEnv0.7.mat')
impulseRecovery
load('30-350HzChirpEnv0.7.mat')
impulseRecovery
s = tf('s'); % Define the Laplace variable
G = 1 / ((s+1)*(s^2 + 2*s + 2)); % Define the transfer function
nyquist(G); % Plot the Nyquist diagram
grid on; % Add grid for better visualization
s = tf('s'); % Define Laplace variable
K = 1; % Given K value
% Forward path
G = K / (s - 1);
% Feedback path
H = (s + 2) / ((s + 1)^2 + 1);
% Open-loop transfer function
L = G * H;
% Closed-loop transfer function
T = L / (1 + L);
% Bode plots
figure;
subplot(2,1,1);
bode(L);
grid on;
title('Bode Plot of Open-Loop Transfer Function L(s)');
subplot(2,1,2);
bode(T);
grid on;
title('Bode Plot of Closed-Loop Transfer Function T(s)');
close all
s = tf('s'); % Define Laplace variable
for K = 1:0.2:2 % Sweep K from 1 to 2 in steps of 0.2
% Forward path
G = K / (s - 1);
% Feedback path
H = (s + 2) / ((s + 1)^2 + 1);
% Open-loop transfer function
L = G * H;
% Compute margins
[GM, PM, wcg, wcp] = margin(L);
% Display results
fprintf('For K = %.1f: Phase Margin = %.2f degrees\n', K, PM);
% Plot Bode and margins
figure;
margin(L);
title(sprintf('Bode Plot with Margins for K = %.1f', K));
```

```
grid on;
end
close all
s = tf('s'); % Define Laplace variable
% Forward path
G = 1 / (s - 1);
% Feedback path
H = (s + 2) / ((s + 1)^2 + 1);
% Open-loop transfer function (K is variable)
L = G * H;
% Root Locus Plot
figure;
rlocus(L);
title('Root Locus of the System');
grid on;
% Find the critical K where the system becomes unstable
[K_critical, poles] = rlocfind(L);
fprintf('Critical K value at stability boundary: %.4f\n', K_critical);
disp('Poles at critical K:');
disp(poles);
close all
close all
s = tf('s'); % Define Laplace variable
K = 1; % Set K to 1 initially
% Forward path
G = K / (s - 1);
% Feedback path
H = (s + 2) / ((s + 1)^2 + 1);
% Open-loop transfer function
L = G * H;
% Nyquist Plot
figure;
nyquist(L);
grid on;
title('Nyquist Plot for Open-Loop Transfer Function L(s)');
%-- 3/7/2025 1:52 PM --%
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 10000);
function [signal, t] = chirpWithEnvelope(freqStart, freqEnd, amplitude, duration, fs)
t = 0:1/fs:duration;
chirpSignal = amplitude * chirp(t, freqStart, duration, freqEnd);
envelope = tukeywin(length(t), min(1, 0.5/duration));
signal = chirpSignal .* envelope;
end
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 10000);
plot(t_chirp, chirpTest, 'r');
close all
invFilter = flip(chirpTest);
plot(invFilter);
length(invFilter)
impulseRecovery
load('30-350HzChirpEnv0.7.mat')
impulseRecovery
length(tempVec{i})(5100:30101))
```

```
length(chirpTest)
impulseRecovery
close all
length(tempVec{i}(5100:30101))
length(tempVec{i}(5100:30100))
length(invFilter)
plot(invFilter)
impulseRecovery
load('30-350HzChirpEnv0.8.mat')
impulseRecovery
load('30-350HzChirpEnv0.8.mat')
y = tempVec{1}(5100:30100)
plot(y)
close all
plot(y)
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 10000);
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 5000);
plot(chirpTest)
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 10000);
plot(chirpTest)
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 5000);
invFilter = flip(chirpTest)
plot(invFilter)
hold on
plot(y)
plot(invFilter)
help conv
hrec = conv(y, invFilter);
plot(hrec)
hrecn = convn(y, invFilter);
hold on;
plot(hrecn)
plot(hrec)
%-- 3/7/2025 2:13 PM --%
load('30-350HzChirpEnv0.8.mat')
[chirpTest, t_chirp] = chirpWithEnvelope(freqStart, freqEnd, amplitude, Tchirp, fs);
[chirpTest, t_chirp] = chirpWithEnvelope(30, 350, 0.8, 5, 5000);
plot(chirpTest)
close all
invFilter = flip(chirpTest)
clc
y = tempVec{1}(5100:30100);
plot(y);
hold on;
plot(invFilt)
plot(invFilter);
close all
yLP = lowpass(y, 1000, 5000);
plot(yLP)
yLP = yLP - mean(yLP);
hold on;
plot(yLP)
close all
```



```
plot(yLP);
order = 700;
cutoffFreq = 11/(5000/2);
b = fir1(order, cutoffFreq, 'high');
yHP = filter(b, 1, yLP);
hold on;
plot(yHP);
order = 500;
b = fir1(order, cutoffFreq, 'high');
yHP = filter(b, 1, yLP);
plot(yHP);
close all
plot(yHP);
chirpTest = chirpTest(249:end);
length(chirpTest);
length(chirpTest)
length(yHP(249:end))
yHP = yHP(249:end);
close all
plot(yHP);
plot(yHP);
hold on;
plot(chirpTest);
invFilter = flip(chirpTest);
hrec = conv(yHP, invFilter);
close all
plot(yHP);
hold on;
plot(chirpTest);
figure;
plot(hrec);
close figure 2;
plot(hrec);
plot(yHP);
close all
plot(yHP);
hold on;
plot(chirpTest);
close all
plot(yHP);
hold on;
plot(invFilter);
figure;
hrec = conv(yHP, invFilter);
plot(hrec);
Hrec = fft(hrec);
nfft = length(H)
nfft = length(Hrec)
Fs = 5000
freq = linspace(0, Fs/2, nfft/2+1);
figure;
plot(freq, abs(Hrec));
freq = linspace(0, Fs/2, nfft);
```

```
plot(freq, abs(Hrec));  
linspace(0, Fs/2, nfft)  
freq = linspace(0, Fs/2, nfft)  
clc  
Hrec= Hrec(1:nfft/2+1);  
plot(freq, abs(Hrec));  
length(Hrec)  
length(freq)  
length(freq)  
freq = linspace(0, Fs/2, nfft)  
freq = linspace(0, Fs/2, nfft/2+1)  
clc  
plot(freq, abs(Hrec));  
hrec_lin = hrec(22800:30600);  
Hrec_lin = fft(hrec_lin);  
figure  
nfft_lin = length(Hrec_lin);  
nfft  
nfft_lin  
Hrec_lin = Hrec_lin(1:nfft_lin/2+1);  
freq_lin = linspace(0, Fs/2, nfft_lin/2+1);  
plot(freq_lin, abs(Hrec_lin));  
close figure 3  
figure 2  
figure 2;  
figure(2);  
close figure 4  
figure  
plot(freq, abs(Hrec));  
hold on;  
plot(freq_lin, abs(Hrec_lin));  
figure  
plot(freq_lin, abs(Hrec_lin));  
close figure 3  
figure  
plot(hrec_lin);  
epsilon  
C = conj(Hrec_lin)./(abs(Hrec_lin).^2+epsilonVec);  
figure;  
plot(epsilonVec)  
epsilon  
plot(epsilonVec)  
C = conj(Hrec_lin)./(abs(Hrec_lin).^2+epsilonVec);  
length(abs(Hrec_lin))  
length(epsilonVec)  
clear epsilonVec  
epsilon  
plot(epsilonVec)  
C = conj(Hrec_lin)./(abs(Hrec_lin).^2+epsilonVec);  
length(C)  
figure  
plot(freq_lin, C);  
plot(freq_lin, abs(C));
```

```
help ifft
cnon = ifft(C);
figure;
plot(cnon);
csym = ifft(C, 'symmetric');
hold on;
plot(csym);
close figure 6;
figure
plot(cnon)
figure;
plot(csym);
length(csym);
length(csym)
length(hrec_lin);
length(hrec_lin)
convImp = conv(hrec_lin, csym);
figure1
figure
plot(convImp);
close all
clear
load('30-350HzChirpEnv0.8.mat')
load('postProcessesing.mat')
clear all
load('postProcessesing.mat')
clearall
clear
load('postProcessesing.mat')
tvec2 = tempVec
plot(tvec2{1})
load('30-350HzChirpEnv0.8.mat')
hold on
plot(tempVec{1})
clear
load('30-350HzChirpEnv0.8.mat')
```